Demonstration of Performing Regression on Datasets

Specifications Required

- Tools: WEKA / Python (pandas, scikit-learn) / Excel
- Dataset: Must contain numeric dependent and independent variables
- Environment: Jupyter Notebook or any Python IDE if using Python
- Basic understanding of regression analysis

Aim

To demonstrate how to perform regression analysis on datasets using tools like Python or WEKA.

Theory

Regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables. The most common type is Linear Regression, which fits a straight line to the data.

Types of Regression:

1. Simple Linear Regression
2. Multiple Linear Regression
3. Polynomial Regression
4. Logistic Regression (for classification)

Steps to Perform Regression in Python

1. Import libraries: pandas, sklearn
2. Load the dataset
3. Split dataset into training and testing sets
4. Fit the regression model
5. Predict and evaluate using metrics like MSE or R2 score

Sample Python Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample data
data = {'X': [1, 2, 3, 4, 5], 'Y': [2, 4, 5, 4, 5]}
df = pd.DataFrame(data)
X = df[['X']]
y = df['Y']

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = LinearRegression()
model.fit(X_train, y_train)
predictions = model.predict(X_test)

# Evaluation
print(mean_squared_error(y_test, predictions))
print(r2_score(y_test, predictions))
```

Conclusion

Regression analysis is a powerful tool for understanding relationships in data. By applying regression techniques using tools like Python or WEKA, we can make accurate predictions and insights.